Troubleshooting

Known gotchas surfaced during design — check here before assuming something's broken.

"Theatre can't reach the mothership at all"

- **Routes not approved.** Advertising a route with `--advertise-routes` isn't enough — it must be approved in the Tailscale admin console (or via `autoApprovers`). Check first; this is the single most common reason a fresh subnet router doesn't work.
- **Check `tailscale status`** on the theatre's A-1300 — confirms whether the peer connection is up at all before debugging further.

"Two theatres can talk to each other" (should be impossible)

- This should never happen if `config/tailscale-acl.json` is loaded correctly — theatres can only reach `tag:mothership`. Confirm the ACL is actually applied in the admin console, not just present in this repo.
- Remember: physical VLAN grouping provides **zero** isolation on its own — it's cabling organization only. If ACLs are missing/misconfigured, cross-theatre reachability is the default, not a bug in the VLANs.

"Tailscale is using DERP relay instead of direct"

- Expected sometimes — see `docs/tailscale.md` for why direct connections are likely but not guaranteed here (depends on whether the Cloud Gateway allows inter-VLAN UDP).
- Run `tailscale netcheck` to see which DERP region is preferred. If it's not the self-hosted `example` region, check: is `derp.example.net` actually resolving? Is the WAN port forward (443/tcp, 3478/udp) actually in place? Neither works until both exist — see `docs/open-questions.md`.
- If the self-hosted DERP is reachable from the mothership's own LAN but not from a theatre, suspect the uplink-VLAN firewall block in `config/unifi/network-config.yaml` catching the hairpin path — see the caveat attached to that rule.

"The NIC bond won't form / only shows one active link"

- **LACP rate mismatch.** Ubiquiti gear hardcodes LACP rate `fast`; Unraid's bonding default is `slow`. Both ends must match — see `docs/topology.md`.
- **Cloud Gateway doesn't support LAG at all** on base (non-Pro/SE/Pro Max) models. If so, bond through an intermediate LACP-capable switch instead — already the agreed fallback, not a new problem.

"BirdDog Play isn't finding the NDI backup source"

- Confirm the Discovery Server IP is actually entered in BirdUI's Network panel on *both* BirdDog Central and every PLAY unit — one-sided config does nothing.
- Plain mDNS will never work across theatres; if Discovery Server isn't configured, this is expected behavior, not a fault. See `docs/streaming-flow.md`.

"Ingested files (ATEM or VMix) aren't showing up in Nextcloud"

- Writing to the folder isn't enough — Nextcloud needs `occ files:scan` to index it. Check the cron job from `setup-nextcloud-external-storage.sh` is actually installed and running, not just printed once during setup.
- If files appear but seem stuck at an old size, check the ingest container's logs — a failed FTP/SMB reconnect will silently stall the pull loop rather than crash it.

"Ingest is falling behind / theatre uplink saturated"

- Confirm the real ATEM bitrate/active-channel count and VMix recording settings — the bandwidth math in `docs/atem-iso-ingest.md` and `docs/vmix-record-ingest.md` assumes figures that were never verified against real hardware.
- If using the `rclone-mount-rsync` alternative, confirm `rsync --append` is actually being used (not falling back to a full re-copy) — check `--itemize-changes` output.

"A copied ATEM/VMix file won't play"

- Confirmed only "very likely" safe to read mid-write, never empirically tested — see the open item in both ingest docs. If it fails, the safest fallback is to only pull once recording has fully stopped (loses the near-real-time property, but guarantees a valid file).

"ATEM Overseer / Flock isn't showing a theatre's live preview"

- Confirm the ATEM/BirdDog Play in that theatre is actually configured to originate its second monitoring/preview stream, not just its primary program feed — this is a genuinely unconfirmed capability, not assumed working out of the box, see `docs/open-questions.md` #11.
- Check the stream is actually reaching the container's IP (`192.168.1.21` for Overseer, `192.168.1.24` for Flock — see `docs/ip-address-map.md`) over the tailnet, same routing path as every other theatre-to-mothership stream in this design.
- If every other theatre works but one doesn't, suspect that theatre's own A-1300 hitting its combined-load ceiling rather than anything Overseer/Flock-side — see the per-router finding in `docs/bandwidth-analysis.md`.

"GL-iNet static DHCP leases aren't applying"

- Every `REPLACE_WITH_MAC_nn` placeholder in `config/gl-inet/` must be swapped for the device's real MAC first — the generated configs ship with placeholders deliberately, not real defaults.